Module-Wrestler Online Character Dictionary

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Abstract

We are the most modern iteration of a long-established translation dictionary relating the language of characters with the language of finite complex group modules.

1 The Translation Dictionary

There are categories **FinGroup** of finite groups, **FinVect** of finite-dimensional vector spaces over $\mathbb{C}$, and for each $G : \mathbf{FinGroup}$, a category $\mathbf{FinVect}^G$ of finite-dimensional complex G-modules. If $V : \mathbf{FinVect}^G$, the G-action will be denoted $G \xrightarrow{\rho_V} \mathrm{GL}(V)$ or $g \cdot x := \rho_V(g)(x)$. If it is clear, we may denote $\rho_V(g)$ by $g \cdot -$. The set of conjugacy classes will be denoted $\mathcal{CC}(G)$. For $x \in G$, its conjugacy class is denoted $\mathcal{CL}(x)$.

1.1 Definitions in the Language of Characters

Throughout, $G : \mathbf{FinGroup}$ is a group and $V : \mathbf{FinVect}^G$ is a finite-dimensional complex G-module. (Any other type data will be specified.)

(D1) **Character** of V: The function $\chi_V := \mathrm{tr} \circ \rho_V : G \to \mathbb{C}$. It is a class function (D3), so it is occasionally factored through the quotient $G \twoheadrightarrow \mathcal{CC}(G)$, where it is regarded as a map $\mathcal{CC}(G) \to \mathbb{C}$.

(D2) **Character table** of G: A listing of the characters χ_I corresponding to the finite-dimensional complex irreducible modules I of G. As characters are class functions (D3), the value on each *conjugacy class* is listed. As the irreducible modules biject with the conjugacy classes, this is a square matrix. For example, behold the character table of S_2 (it's not interesting):

Character	$(*)(*)$	$(**)$
χ_{trivial}	1	1
χ_{sign}	1	-1

Beware – although the row for χ_I encodes the character χ_I, it is missing information on the size of the conjugacy class. This information is critical to computing the inner product of two characters (D4); generically, the inner product **does not** agree with the "dot product".

(D3) **Class function**: A function $\phi : G \to A$ which is constant on conjugacy classes of G. A is typically $\mathbb{C}$.

(D4) **Inner product** on $\mathbb{C}^G$: The vector space of functions $G \to \mathbb{C}$ carries Hermitian inner product $\langle -, - \rangle$ given by

$$\langle \phi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}$$

or, for characters $(U, V : \mathbf{FinVect}^G)$:

$$\langle \chi_U, \chi_V \rangle := \frac{1}{|G|} \sum_{g \in G} \chi_U(g) \cdot \chi_V(g^{-1})$$

because of (G1).

1.2 Grammatical Rules of the Language of Characters

I mean in this section to list some key properties that characters always satisfy. If you were to communicate using characters, these are some rules that "always appear" in your sentences! Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$ unless otherwise specified.

(G1) **Inverses**: For $g \in G$, $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

(G2) **Frobenius reciprocity**: If $H \leq G$, $U : \mathbf{FinVect}^H$ and $V : \mathbf{FinVect}^G$, then

$$\langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

See (T1).

(G3) **Orthonormality of irreducibles**:

$$U, V \text{ both irreducible} \implies \langle \chi_U, \chi_V \rangle = \mathbf{1}_{U \simeq V} \quad \text{(indicator function)}.$$

1.3 Translating Between Characters and Representations

Throughout, $G : \mathbf{FinGroup}$ and $U, V : \mathbf{FinVect}^G$.

(T1) **Adjunction of induction to restriction**: For a subgroup $H \leq G$, the adjunction of functors $\mathrm{Ind}_H^G \dashv \mathrm{Res}_H^G$ appears as **Frobenius reciprocity**:

$$\mathbf{FinVect}^G(\mathrm{Ind}_H^G U, V) \simeq_{\mathrm{nat}.U,V} \mathbf{FinVect}^H(U, \mathrm{Res}_H^G V) \quad \rightsquigarrow \quad \langle \mathrm{Ind}_H^G U, V \rangle = \langle U, \mathrm{Res}_H^G V \rangle.$$

The inner product relation is a decategorification of the (categorical) adjunction.

(T2) **Dimension/Degree** of V:

$$\dim(V) = \chi_V(1).$$

(T3) **Dimension** of hom-space $\mathbf{FinVect}^G(U, V)$ (this space goes by the notation $\mathrm{Hom}_G(U, V)$ in the course):

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle.$$

(T4) **Equivalence of irreducible modules**: In the context where U, V are irreducible,

$$U \simeq V \iff \langle \chi_U, \chi_V \rangle = 1$$

and furthermore, in the same context,

$$U \not\simeq V \iff \langle \chi_U, \chi_V \rangle = 0.$$

See (G3) for a more concise statement.

(T5) **Frobenius reciprocity**: See (T1).

(T6) **Irreducible factors**: For U irreducible,

$$\text{a factor of } U \text{ appears in } V \iff \langle \chi_U, \chi_V \rangle \neq 0.$$

(A factor "appearing" means that $V \simeq U \oplus (\text{stuff})$.)

(T7) **Irreducibility**: See also (G3).

$$V \text{ is irreducible} \iff \langle \chi_V, \chi_V \rangle = 1.$$

2 The Story of Characters, Told in Broad Strokes

If you want, this section charts the "etymology" of each word in the dictionary!

The slogan is that **characters decategorify modules**, in analogy to how **dimension decategorifies vector spaces**.

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{character}} & \text{Decategorified} \qquad\qquad \text{What is lost?} \\ \\ \text{Module } U & \longmapsto & \text{List of numbers } \chi_U \in \mathbb{C}^G \qquad \text{Algebraic data (action } G \to \mathrm{GL}(U)) \\ \Big\downarrow{\scriptstyle \text{morphism space}} & & \Big\downarrow{\scriptstyle \text{its dimension}} \qquad\qquad \Big\downarrow{\scriptstyle \text{transformation of maps}} \\ \text{Module } V & \longmapsto & \text{List of numbers } \chi_V \in \mathbb{C}^G \qquad \text{Algebraic data: (action } G \to \mathrm{GL}(V)) \end{array}$$

To exemplify the analogy:

$$\begin{array}{ccc} \text{Categorified} & \xrightarrow{\text{dimension}} & \text{Decategorified} \qquad\qquad \text{What is lost?} \\ \\ \text{Vect. Sp. } U & \longmapsto & \text{Number } \dim U \in \mathbb{Z}_{\geq 0} \qquad \text{Algebraic data (vectors)} \\ \Big\downarrow{\scriptstyle \text{morphism space}} & & \Big\downarrow{\scriptstyle \text{rank-nullity statements}} \qquad\qquad \Big\downarrow{\scriptstyle \text{transformation of algebraic data}} \\ \text{Vect. Sp. } V & \longmapsto & \text{Number } \dim V \in \mathbb{Z}_{\geq 0} \qquad \text{Algebraic data (vectors)} \end{array}$$

In both situations, **the decategorification classifies the categorification up to isomorphism** – that is, two (finite-dimensional) vector spaces are isomorphic just when they have the same dimension, and two (finite-dimensional, complex) G-modules are isomorphic just when they have the same character. At the level of objects, the decategorification retains all necessary information (as one would hope of anything being granted the monacre of "decategorification"). However, passing to this "more computational invariant" has the deficiency of losing algebraic data. On decategorifying **FinVect**, we lose large amounts of data on how linear maps put together vector spaces. On decategorifying $\mathbf{FinVect}^G$, we lose large amounts of data on the actual action of G, and we lose information on how these actions intertwine.

2.1 Prologue: The Birth of Characters

We now tell a fiction on the story of character theory, all in broad strokes. Once upon a time, vector spaces were decategorified by their dimension, and everyone lived happily ever after. Well, they lived up until algebraists decided to represent symmetries on vector spaces, moving us from the land **FinVect** to the various lands $\mathbf{FinVect}^G$ (for $G : \mathbf{FinGroup}$). In early and tumultuous days, many discoveries were made paralleling the story for vector spaces:

	Vector spaces	Modules
can be broken up into	one-dimensional subspaces,	irreducible submodules;
the maps between which are	effectively scalars,	effectively scalars;
and which form all objects by	direct sum,	direct sum.

After these initial algebraic gains, however, lingered important questions – "How many irreducible modules are there?", or "Where can we find these modules?", for example. To gain further ground, the category $\mathbf{FinVect}^G$ needed to be quite seriously tamed. After dusting off the history books, the algebraists attempted to decategorify $\mathbf{FinVect}^G$ by mirroring the dimension decategorification of **FinVect**. Fortunately, in the context of a group representation, the dimension appears

naturally:

$$G \xrightarrow{\rho_V} \mathrm{GL}(V) \quad \rightsquigarrow \quad \rho_V(1_G) = 1_V = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}}_{n \times n} \quad \rightsquigarrow \quad \dim(V) = n = \mathrm{tr}(\rho_V(1_G)).$$

This is a natural choice because the representing linear transformations $\rho_V(g), g \in G$ are the only extra data present within objects of $\mathbf{FinVect}^G$, and the only such transformation guaranteed to exist is $\rho_V(1_G) = 1_V$. The generalisation easily revealed itself – the **character** of V is the function $\chi_V : G \to \mathbb{C}$ given by $\chi_V := \mathrm{tr} \circ \rho_V$. The miracle is that this successfully decategorified away enough complexity to make tractable progress in understanding the category $\mathbf{FinVect}^G$.

2.2 Chapter One: Separating Modules with Inner Products

The preceding text is best understood as a prologue to our story – "Chapter 0", if you will. In chapter one, we begin to understand the hom-spaces $\mathbf{FinVect}^G(U, V)$ for $U, V : \mathbf{FinVect}^G$. Every part of our story takes place over the category of vector spaces, so it is not a surprise to find that $\mathbf{FinVect}^G(U, V)$ is a vector space. Recording its dimension would be a first feat, and no easy one at that. In perhaps a stroke of luck, or perhaps by arcane insight dug up from historical texts[1], one can find some inspiration in a superficial look at $\mathbf{FinVect}^G(-, -)$:

$$\begin{array}{lll} \text{Categorified} & \mathbf{FinVect}^G \times \mathbf{FinVect}^G \longrightarrow \mathbf{FinVect} & \\ & (U, V) \longmapsto \mathbf{FinVect}^G(U, V), & \\ \text{character} \downarrow & & \\ \text{Decategorified} & \mathbb{C}^G \times \mathbb{C}^G \dashrightarrow \mathbb{C} & \\ & (\chi_U, \chi_V) \longmapsto ?. & \end{array}$$

The superficial symmetry is that $\mathbf{FinVect}^G(-, -)$ appears to decategorify into something resembling an inner product. There is a natural choice of inner product on the set of functions $G \to \mathbb{C}$, given by

$$\begin{aligned} \mathbb{C}^G \times \mathbb{C}^G &\xrightarrow{\langle -, - \rangle} \mathbb{C} \\ (\phi, \psi) &\longmapsto \frac{1}{|G|} \sum_{g \in G} \phi(g) \cdot \overline{\psi(g)}. \end{aligned}$$

The division by $|G|$ may be seen as a normalisation factor; this had proven useful to the algebraists prior. The miracle of chapter one is that *this inner product idea worked* and gave rise to a rich theory, eventually giving the dimension formula

$$\dim \mathbf{FinVect}^G(U, V) = \langle \chi_U, \chi_V \rangle$$

and answering some of the lingering questions prior. How these results follow is not immediate, and will take us another chapter to describe.

2.3 Chapter Two: Orthonormality of Irreducible Characters

A first observation is that

[1] The *Module-Wrestler Online Character Dictionary* keeps no good record of history.

Lemma 2.1 (Characters of direct sums decompose as sums of characters)
Let $\{V_\alpha\}_{\alpha\in\mathcal{A}}$ be objects of $\mathbf{FinVect}^G$. Then, $\chi_{\oplus_{\alpha\in\mathcal{A}}V_\alpha} = \sum_{\alpha\in\mathcal{A}}\chi_{V_\alpha}$.

Since any module decomposes uniquely into a direct sum of irreducible submodules, we should first study the characters χ_I where I is irreducible. If our inner product $\langle -,-\rangle$ really is to count dimensions of hom-spaces, then Schur's lemma tells us we should expect that

$$\begin{aligned} U \simeq V \text{ are irreducible } &\implies \mathbf{FinVect}^G(U,V) \simeq_{\mathbf{FinVect}} \mathbb{C}^1 \implies \langle\chi_U,\chi_V\rangle = 1, \\ U \not\simeq V \text{ are irreducible } &\implies \mathbf{FinVect}^G(U,V) \simeq_{\mathbf{FinVect}} 0 \implies \langle\chi_U,\chi_V\rangle = 0, \end{aligned}$$

which is to say in other words that if we enumerate the characters as χ_I, one for each isomorphism class of irreducible modules $I : \mathbf{FinVect}^G$, then these characters are orthonormal. Sure enough,

Proposition 2.2 (Orthogonality of irreducible characters)
For irreducible modules $I, J : \mathbf{FinVect}^G$,

$$\langle\chi_I,\chi_J\rangle = \mathbf{1}_{I\simeq J},$$

with the $\mathbf{1}$ denoting an indicator function. (Perhaps think of the Kronecker delta.)

Now, let us collect together one irreducible G-module from each isomorphism class into a set $\mathcal{I}(G)$ (ignore size issues for now) and write two direct sum decompositions $U \simeq \bigoplus_{I\in\mathcal{I}(G)} I^{\oplus u_I}$ and $V \simeq \bigoplus_{I\in\mathcal{I}(G)} I^{\oplus v_I}$. Considering the universal properties of the direct sum, Schur's lemma tells us that

$$\begin{aligned} \mathbf{FinVect}^G(U,V) &\simeq \bigoplus_{I,J\in\mathcal{I}(G)} \mathbf{FinVect}^G(I,J)^{\oplus u_I v_J} \\ &\simeq \bigoplus_{I,J\in\mathcal{I}(G)} \left(\mathbb{C}^{\mathbf{1}_{I=J}}\right)^{\oplus u_I v_J} \\ \mathbf{FinVect}^G(U,V) &\simeq \bigoplus_{I\in\mathcal{I}(G)} \mathbb{C}^{\oplus u_I v_I} \\ \implies \quad \dim \mathbf{FinVect}^G(U,V) &= \sum_{I\in\mathcal{I}(G)} u_I v_I \end{aligned}$$

while if we compute the characters, the previous proposition shows us that

$$\begin{aligned} \langle\chi_U,\chi_V\rangle &= \sum_{I,J\in\mathcal{I}(G)} u_I v_J \langle\chi_I,\chi_J\rangle \\ &= \sum_{I,J\in\mathcal{I}(G)} u_I v_J \mathbf{1}_{I=J} \\ \langle\chi_U,\chi_V\rangle &= \sum_{I\in\mathcal{I}(G)} u_I v_I \end{aligned}$$

and therefore,

Theorem 2.3 (Inner products of characters decategorify hom-spaces)
For $U, V : \mathbf{FinVect}^G$, we have

$$\langle\chi_U,\chi_V\rangle = \dim \mathbf{FinVect}^G(U,V).$$

What we have seen also gives us a strategy to extract from a character $\chi_V = \sum_{i\in\mathcal{I}(G)} a_i \cdot \chi_{V_i}$ the coefficients $a_i \in \mathbb{Z}_{\geq 0}$ declaring the "multiplicities" in the direct sum $V \simeq \bigoplus_{i\in\mathcal{I}(G)} V_i^{\oplus a_i}$ of irreducible modules V_i. One can either leverage orthonormality of the irreducible characters (to extract the coefficients straight away), or leverage the fact that $a_i = \dim \mathbf{FinVect}^G(V_i, V)$ (which is not hard to prove). Either way,

Proposition 2.4 (Irreducible characters give an orthonormal basis for all characters)
Just as the irreducible submodules give a "direct sum basis" for the finite-dimensional complex G-modules up to isomorphism, the irreducible characters give a "$\mathbb{Z}_{\geq 0}$-weighted basis" for their characters. That is, we have a correspondence

$$\mathbf{FinVect}_{/\simeq}^{G} \longleftrightarrow \{\text{characters of } G\}$$
$$V \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus a_I} \longleftrightarrow \chi_V = \sum_{I \in \mathcal{I}(G)} a_I \cdot \chi_I$$

where $\mathcal{I}(G)$ collects each isomorphism class of irreducible G-modules. (Ignore, for now, the worry that constructing $\mathcal{I}(G)$ may seem to ask for a choice function over proper classes. We will later provide a recipe in the language of group representations ot construct $\mathcal{I}(G)$.)

The previous proposition has a linear algebraic deficiency. Part of the statement is that the irreducible characters give a "basis" of the set of all characters. It would be nice to interpret this as an actual basis for a vector space, but the set of characters is not closed under scaling by $\mathbb{C}$. Rather than doing something artificial (e.g. taking the $\mathbb{C}$-span of this set), we can consider the vector space $\mathbb{C}^G$. Thinking categorically, we find that

Lemma 2.5 (Characters are class functions)
Just as conjugate (isomorphic) elements x and gxg^{-1} of a group have conjugate (isomorphic) representations $\rho_V(x)$ and $\rho_V(g)\rho_V(x)\rho_V(g)^{-1}$, the character assigns these elements equal values. That is, $\chi_V(x) = \chi_V(gxg^{-1})$. (This decategorifies isomorphism into equality.)

This does mean that characters cannot span $\mathbb{C}^G$ for non-Abelian G. However, it also shows us a better target for their span.

Theorem 2.6 (Irreducible characters are an orthonormal basis for class functions)
The set $\left\{\chi_I \;\middle|\; I : \mathbf{FinVect}^G \text{ is irreducible}\right\}$ is an orthonormal basis for $\mathbb{C}^{\mathcal{CC}(G)}$. There is hence one irreducible character of G for each conjugacy class of G. Categorifying, there is (up to isomorphism) one irreducible module of G for each conjugacy class of G.

We can now conclude that the class $\mathcal{I}(G)$ is finite, though it remains to be shown that it can be constructed as a set.

2.4 Chapter Three: Characters, Conjugacy and the Left-Regular Representation

All good things come in threes, and the missing partner in our story is the left-regular representation $\mathbb{C}[G]$. A telling of this story which does not take such broad strokes would actually reveal, in showing that the irreducible characters span $\mathbb{C}^{\mathcal{CC}(G)}$ (as part of proving 2.6), that the left-regular representation appears. To take yet another broad stroke, the conclusion here is that this representation can be completely torn apart.

Proposition 2.7 (The left-regular representation summarises the irreducible modules)
The left-regular representation $\mathbb{C}[G]$ decomposes as

$$\mathbb{C}[G] \simeq \bigoplus_{I \in \mathcal{I}(G)} I^{\oplus \dim I},$$

or in the language of characters,

$$\chi_{\mathbb{C}[G]} = \sum_{I \in \mathcal{I}(G)} \dim I \cdot \chi_I.$$

We can thus construct $\mathcal{I}(G)$ as a set, as each irreducible module occurs in the *set* of submodules of $\mathbb{C}[G]$. Furthermore, the centre of the endomorphism algebra captures the conjugacy classes and irreducible modules together:

$$|\mathcal{CC}(G)| = \dim\left(Z\left(\mathbf{FinVect}^G(\mathbb{C}[G], \mathbb{C}[G])\right)\right) = |\mathcal{I}(G)|.$$

This gives another proof, independent of 2.6.

2.5 Chapter Four: Induction is Left Adjoint to Restriction

FIXME:

3 Proofs

We collect together proofs of all statements in the translation dictionary and in the "story told in broad strokes" (sections 1 and 2).

4 Some Character Tables